



NonStop
Innovation

FUTURE-PROOFING NONSTOP

Phil Ly

A Stroll Down Memory Lane



Happy 50th
Anniversary
to Tandem
Computers

1974 - 2024



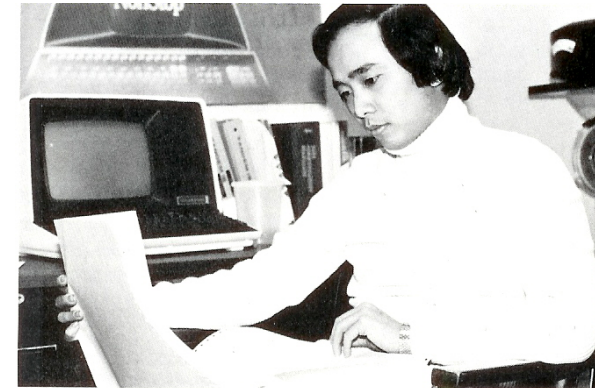
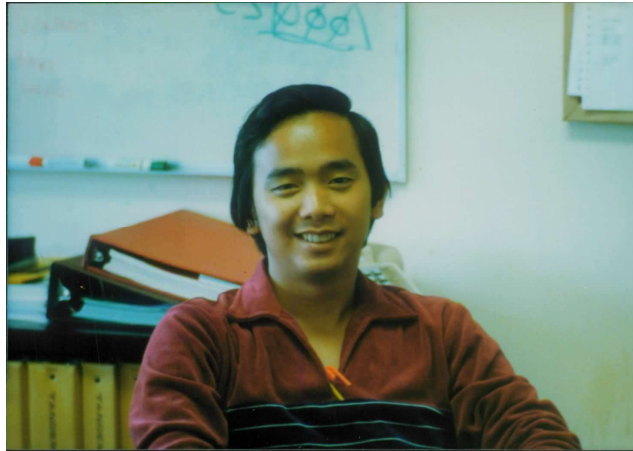
1974 – Where were you?

Internship at IBM, Kingston



My Tandem Journey

1979 - 1983



Phil Ly

TANDEM

PHIL LY
Manager Software Development

TANDEM COMPUTERS INCORPORATED
12100 Sunrise Valley Drive
Reston, VA 22091
(703) 476-0550

TANDEM
COMPUTERS

PHILIP LY
Regional Datacommunications Specialist

(212) 760-8440
One Penn Plaza • New York, New York 10001

Lessons from Tandem Computers



EMAIL ID
COMMUNICATION



LOGON ID
HANDS ON



TANDEM EDUCATION
CONTINUOUS
LEARNING

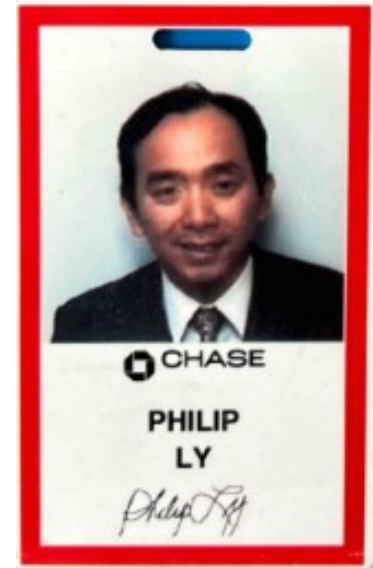


BEER BUST
COLLABORATION



CUSTOMER FIRST
SERVICE

TIC's Journey Began in 1983



2000
An Amazing
Year
at TIC



Future Proofing NonStop

fu·ture-proof

/ˈfyʊtʃərˌpruːf/ 

BRITISH

adjective

1. (of a product) unlikely to become obsolete.

verb

1. make (a product or system) future-proof.
"this approach allows you to future-proof your applications"



Success Oriented Services



TIC Software's Tenets



Customers First

Share knowledge

Give value

Think solutions



Continuous learning

Keep up with modern technology

Integrate

Think outside the box



Collaboration

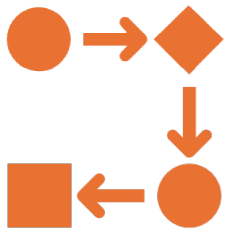
Customers

HPE

Partners

NonStop Community

How to Future-Proof NonStop



Modernize

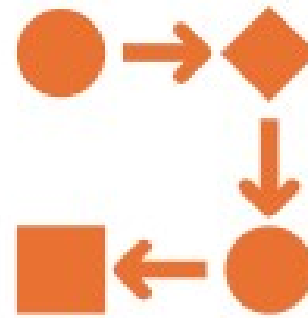


Explore New
Technology



Invest in people

Future
Proof
NonStop



Modernize



Enterprise

Google™



amazon

Internet

LightWave
Server



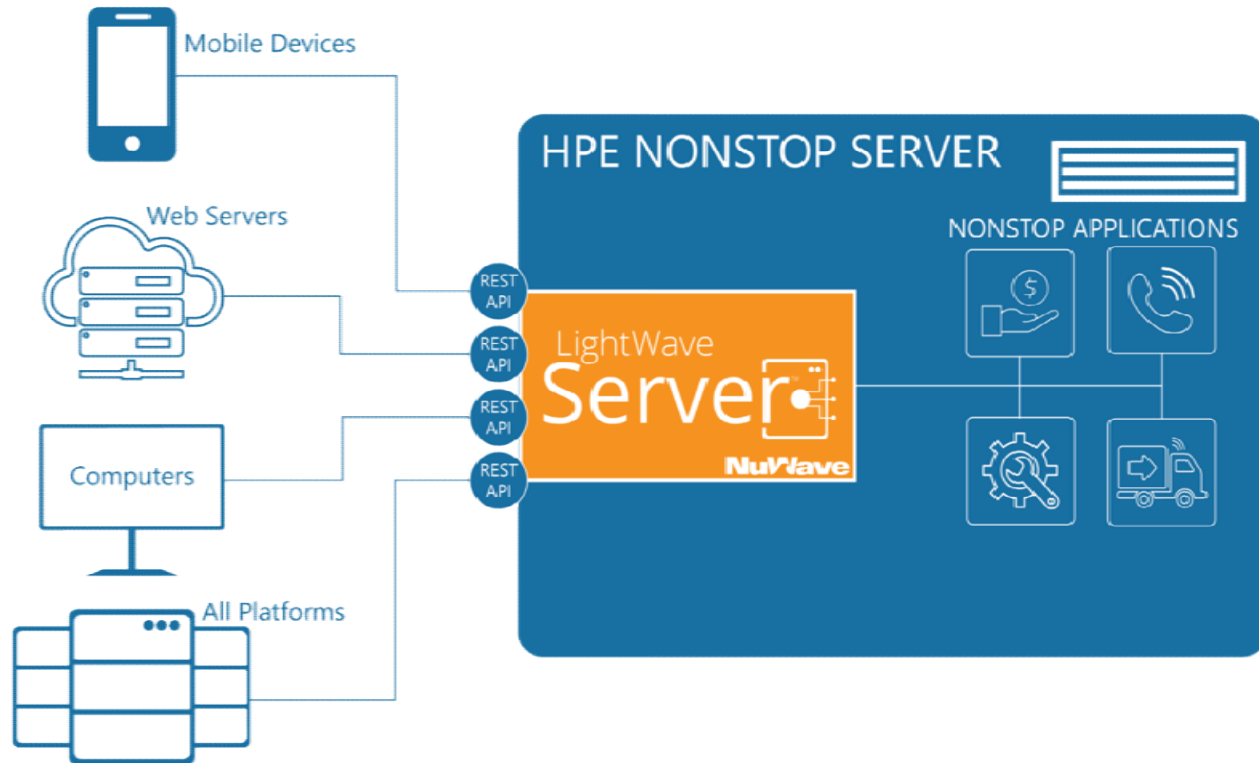
NonStop



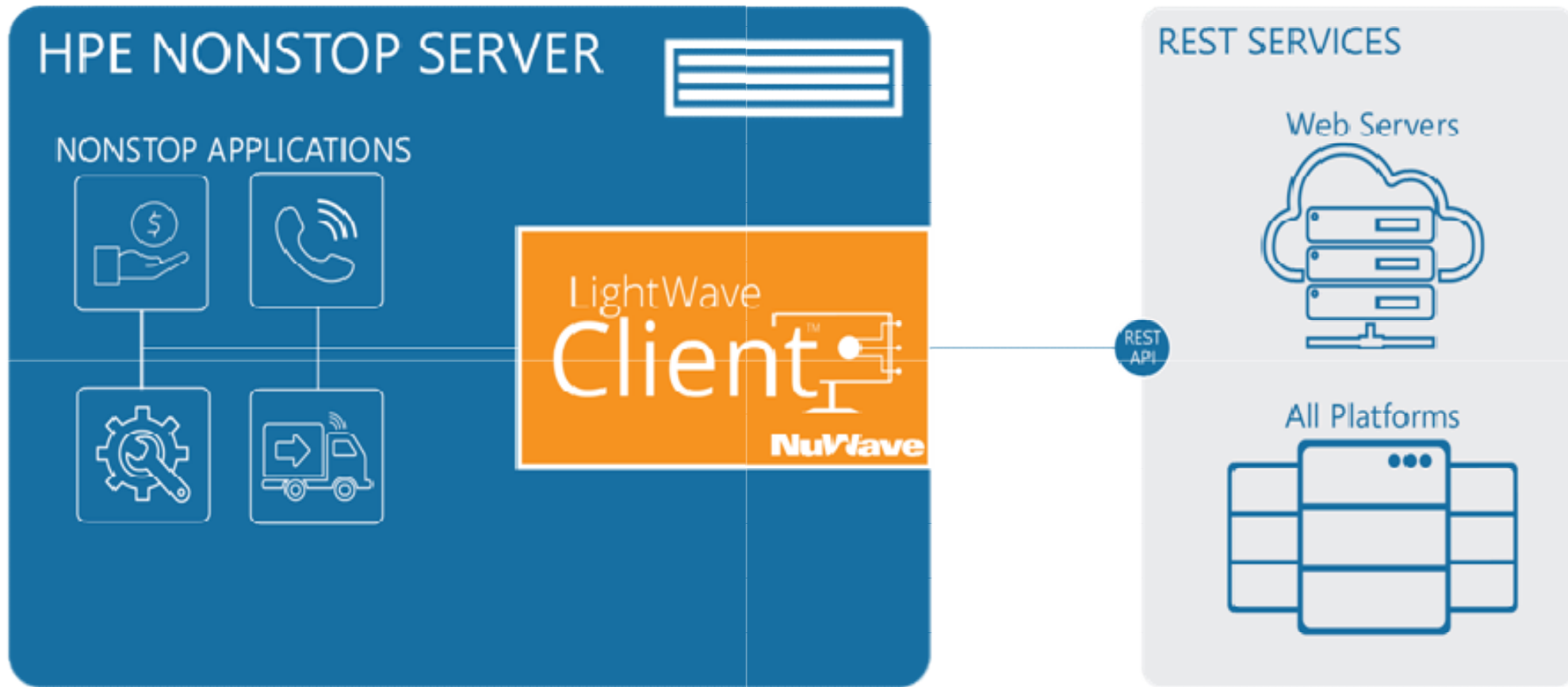
LightWave
Client

 **MuleSoft**
API Gateway

LIGHTWAVE SERVER™



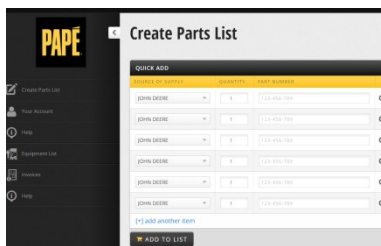
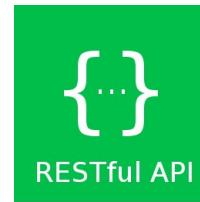
LightWave Client™



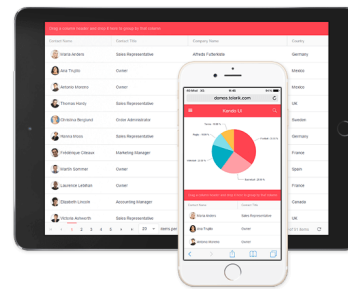
Modernize UI



NonStop



Browser



Mobile

Green Screen to Browser

```
ADS - TMS / G10          THE PAPER GROUP      5/13/20 12:25:52
STNO.#U1488              ACCOUNTING DEPARTMENT 255,255

AP6      AP INQUIRY DISP/REP      GL21      JD/HYST/DW/FC INQ
FILEPERO FILE/DISK % REPORT      TICTEST   TIC TEST MENU

JOHNCONV JOHNSON LIFT CONVSNS    HPP304    RENTAL INVCS INQUIRY

SQLTEST  SQL TESTING            HPROD14   SHIP TO FILE MAIN.
DSQLTEST DW SQL TESTING
USQLTEST PMAC SQL TESTING
KSQLTEST KW SQL TESTING        LUANN     LUANN DEFINE TEST
HSQLTEST PMH SQL TESTING        LCRTEST   ENSCRIBE DEFINE

JTPP514  JD WHOLE GDS INV TST
HTPP514  HY WHOLE GDS INV TST    KPROD10   P/T PARTS/MKTG INQRY
KTPP514  KW WHOLE GDS INV TST
DTPP514  DW WHOLE GDS INV TST    AMTEST    AMTEST MENU

ENTER SELECTION -->
```

LightWave ServerTM

Web Reporting | Papé Remote Support | TurboMeeting | Log Out

Company: 02 - Papé Machinery C&F | Branch: All Branches (00 - EUGENE, 01 - PORTLAND, 02 - FERNBRIDGE) | Customer Number: Customer # | Customer Name: A | Search Type: Contain | Inc.Deleted: No

Address 1: Address 1 | Address 2: Address 2 | City: City | Phone Number: 541 999 9999 | Search | Excel

We have limited the results to 500. Please narrow your search.

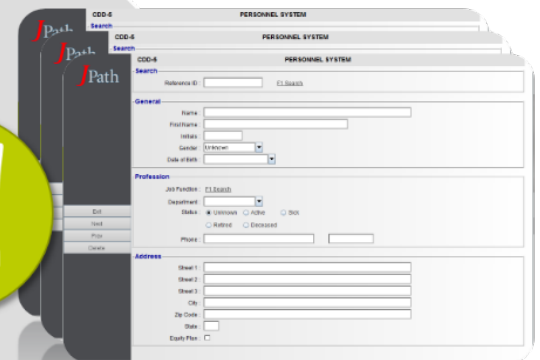
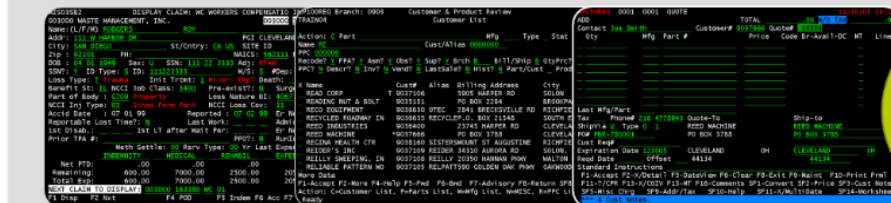
Show 10 entries

CO	BR	Cust_#	Cust_Name	Address_1	Address_2	City	State	Phone	Remaining_Credit	Terms
2	2	493533	(JD) AARON PAINE	MOBILE WRENCHING	82 VALLEY AVE	FORTUNA	CA	707-502-5397	\$1,107.97	
2	2	492704	(JD) ADAM TAIT	PO BOX 511		FERNDAL	CA	707-292-0720	\$10,000.00	
2	2	490005	(JD) ALEXANDRE ACRES	8371 LOWER LAKE ROAD		CRESCENT CITY	CA	707-786-4373	\$2,183.64	COD
2	2	490004	(JD) ALEXANDRE DAIRY	8371 LOWER LAKE ROAD		CRESCENT CITY	CA	707-951-1000	\$13,382.31	
2	2	5289849	(JD) ARLEY SMITH	366 ROCK PIT ROAD		MCKINLEYVILLE	CA	707-339-1217	\$1,000.00	
2	2	490593	(JD) ART R TOWNSEND	PO BOX 1174		FERNDAL	CA	707-786-9125	\$1,000.00	
2	2	490232	(JD) BAR M RANCH	3364 FOSTER AVENUE		ARCATA	CA	707-496-0609	\$1,320.06	

Transforming Pathway Screens

JPath

Providing a modern graphical user interface for HPE NonStop 6530 and IBM 3270 applications



From Overcrowded Green Screens . . .

JPath

Providing a modern graphical user interface for HPE NonStop 6530 and IBM 3270 applications

```
CLS14SB2          CHANGE CLAIM: WC WORKERS COMPENSATION IND OP 24-Sep-2012 A
001404 THE PAUL(R) KEIPERT CORPORATION          001404 000156 WC 01 00 001
Name: (L/F) STORM          WAYNE          | | TEST CENTER ONE
Addr: 890 LILLY LANE          | | PROFIT CENTER : PROFITCTR1
City: PALATINE          St/Cntry: IL US          | | FEIN: 28 1094818 SIC: 0116
Zip : 60333 4321 Phone: 847/285-3743 X3222          | | NAICS: 112112 MLO:
DOB : 01 18 1862 Sex: F SSN: 999 01 0001 Adj: KEIPERT Br: ITASCA,IL
SSN?: N ID Type: G ID: 997352556 M/S: S #Dep: 05 Exposure: UT US
Loss Type: C Cun. Inj. Init Trtmt: 4 Hosp > 24 h Death: from inj:
Benefit St: NE NCCI Job Class: 0400 Pre-exist?: N Surgery?: N / 06 01 89 -
Part of Body : 0002 Abdomen Loss Nature BI: 9047 Alle
NCCI Inj Type: 09 PERM PARTIAL NCCI Loss Cov: 11 STAT
Accid Date : 01 08 81 Reported : 04 15 81 Er Notif
Reportable Lost Time?: Y Last Work: 01 08 81 Er Notif
1st Disab.: 01 08 81 1st LT after Wait Per: 01 13 83 Admin Not
Indxd: 05 10 10 Meth Settle: 03 Rsrv Type: 00 Yr Last Expsd:
      INDEMNITY MEDICAL REHABIL EXPENSE
Net PTD: 120.00 6.00 2.00 42.00
Remaining: 80.00 94.00 99.00 4556.00
Total Exp: 200 100 100 4598
NEXT CLAIM TO CHANGE: 001404 000156 WC 01
F1 Dsp F2 Nxt F4 POD F7 TEW F8 Emp SF2 Index S
```

```
CLS14SB3          CHANGE CLAIM: WC WORKERS COMPENSATION IND OP 24-Sep-20
Client: 001404 THE PAUL(R) KEIPERT CORPORATION          001404 000156 WC 01 00
Name: (L/F) STORM          WAYNE
=== EMPLOYMENT INFO === NCCI Job Class: 0400 COTTON COMPRESSING & DRIVER
Hire Date: 07 17 64 Status: S ON STRIKE Workdays/Wk
Occupation: RADIOLOGIST...+...2...+...3...+...4...+...5 Len of Serv
Department: RADIOLOGY
Full Pay Day Inj?: N Salary Cont/Lieu Benef?: N Period: - - -
WEEKLY WAGES Avg: .67 Effect Date: 09 22 11 Comp Rate: 7
Other Comp: 12.03 Post-Injury: 40.00 Agree To Compensat
===== MISC INFO ===== Prior TPA #: OP28932-490-92352342
RECOV POTENTIAL -Subro?: N -St Fund?: N St Claim #: STATE CLAIM NUMBER: X
BENEFIT OFFSETS -Unemploy?: N -Pension?: N [SSN?: N Other?: N Amt:
Attny/Auth Rep(F,Y,N) & Disclosure Date: N / Date of Rep: 07 0
Controverted?: Y Denial Date: Legal Firm: 000000 Indep Adj: 99
In Suit?: Y Deductible?: N Fraud: N Lien?: Y
Claim Descr: DESCRIPTION - SETUP WC FROM RISX-FACS MENU
SHORTER UPDATES ARE OKAY. UPDATE .... UPDATE ...
Inj. Desc:
(NY Only)
Optional Codes: 1. NEW 3RD CO 2. 3. 4.
NEXT CLAIM TO CHANGE: 001404 000156 WC 01
F1 Dsp SF4 Chg SF5 Supp SF6 Descr
```



JPath

Beautiful GUI

JPath

Providing a modern graphical user interface for HPE NonStop 6530 and IBM 3270 applications

CLS145B2 CHANGE CLAIM: WC WORKERS COMPENSATION IND OP 06-Jan-2016 A
003000 WASTE MANAGEMENT, INC. 003000-163380-WC-01 00 001
Adjuster: RFAW Accident Date: 07 01 99 Branch: ITASCA IS T

Next Claim to Change : 003000 163380 WC 01 Go

Reporting Unit
PGI CLEVELAND
SITE ID : 6625

Claimant
Last Name : RODGERS First : ROY
Address : 111 W HARBOR DR
City : SAN DIEGO
State : CA Postal Code : 92101 Country : US
Phone :

Claimant Identifier
Does Claimant Have a SSN : Yes
ID : 111223333 ID Type :
SSN : 111 22 3333
Gender : Unknown
Date of Birth : 04 01 1949
Date of Death :
Marital Status : Single, divorced, widowed Result of Injury :
Number of Dependents :

Injury
Reported Date : 07 02 99
Benefit State : IL
Exposure State :
Nature(BI) : 4067 Other
Part of Body : G709 Property
Injury Type : 03 Sched Perm Part
Type of Loss : Trauma
Loss Coverage : 11 STATE ACT TRAUMA
Initial Treatment : Minor on-site remedies by employer medical staff
Surgery : No Date of Surgery :
Pre-existing Condition : No
PPO : No
MMI Date :
% Impairment :

Legal
Fraudulent : Not Fraudulent
Claim in Suit : No
Claimant Represented : No Attorney or Authorized Representative
Legal Firm Number :
Lien : No
Indicates whether or not the claimant has an attorney or authorized representative. For WC it also defines whether or not the Attorney Disclosure Form has been filed. Workers's Compensation (except NJ) Definition: F = Yes, Attorney Disclosure Form filed. Y = Yes, but no Attorney Disclosure Form filed. N = No Attorney or Authorized Representative.

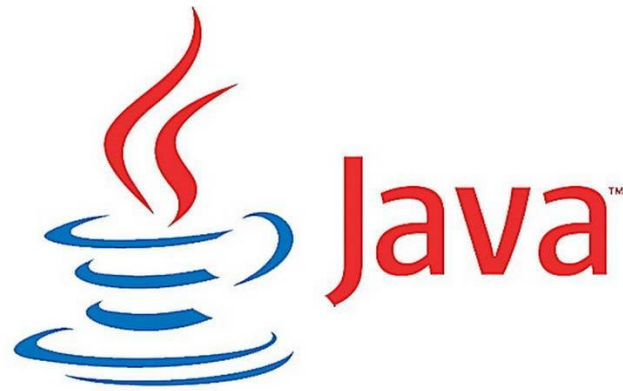
Optional Codes
1. 2. 3. 9999999999

Administration
Doctor's Restrictions and Employer's Ability to Accommodate
Restricted Return to Work
Date Employee Returned to Work : 01 14 12
Restrictions : Temporary
Same Employer : Yes Same Position : No
Same Duties : No
Unrestricted Return to Work (Full Duty)
Date Employee Returned to Work :
Reason :
Doctor's Release with No Return to Work
Date Employee Not Returned to Work :
Reason :
Death
Disagree with doctor
Disagree with job duty
Position not available
Terminated
Transportation
Unable to modify duty



JPath

Modernized Development

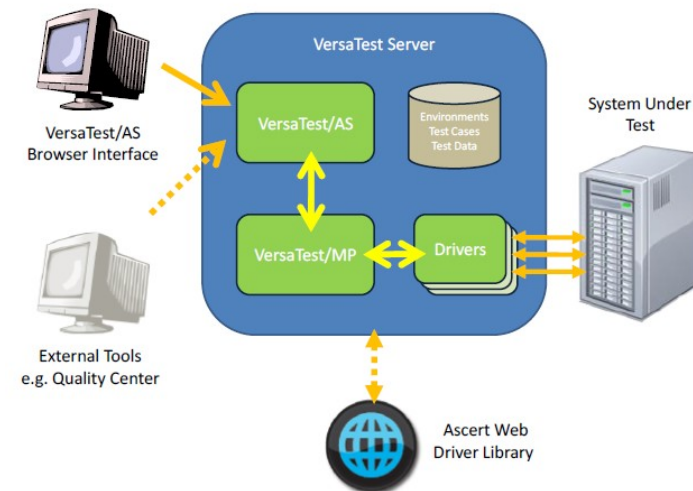
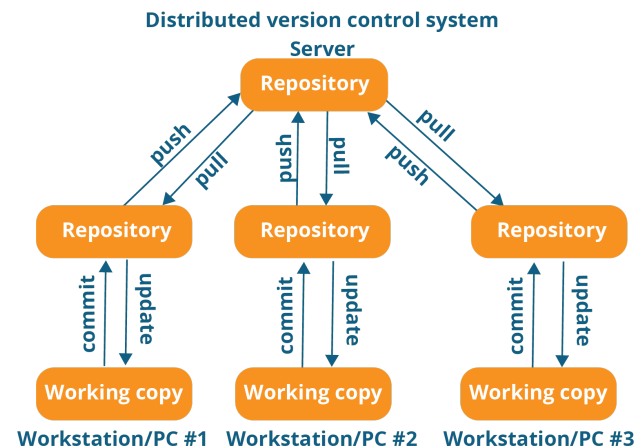


Modern Productivity Tools



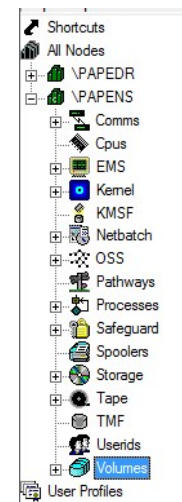
NSDEE

Industry standard
Intellisense editing
Desktop compilation
Easy online
debugging



TOP GUI Tool for NonStop

- The Operation Pack (TOP)
 - Communications
 - CPU
 - EMS
 - Netbatch
 - Pathway
 - Safeguard
 - Spooler
 - Many, many more...

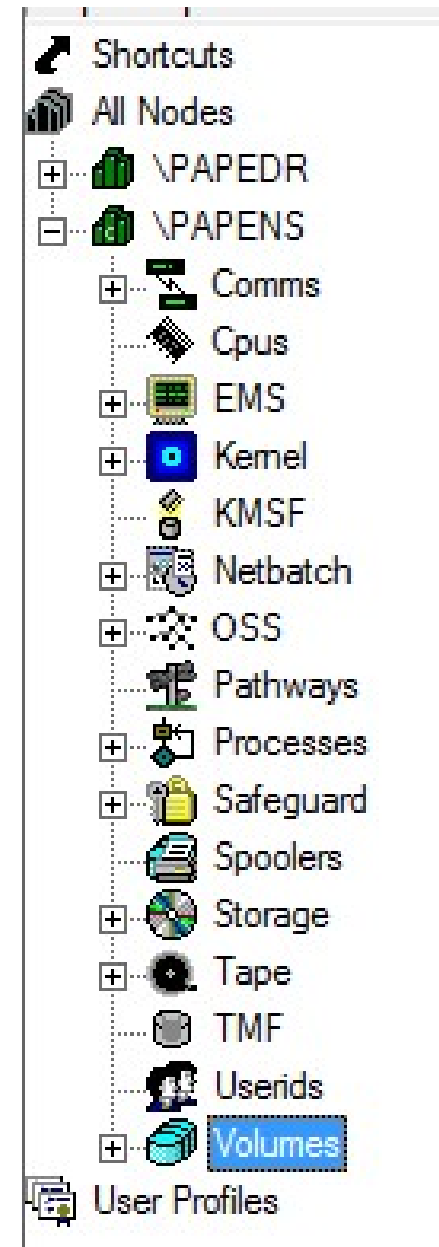


TOP Benefits

TOP has a **complete** NonStop operation interface

Windows-like UI makes TOP easier for new staff to learn NonStop concepts

TOP can also be a great **collaboration tool**

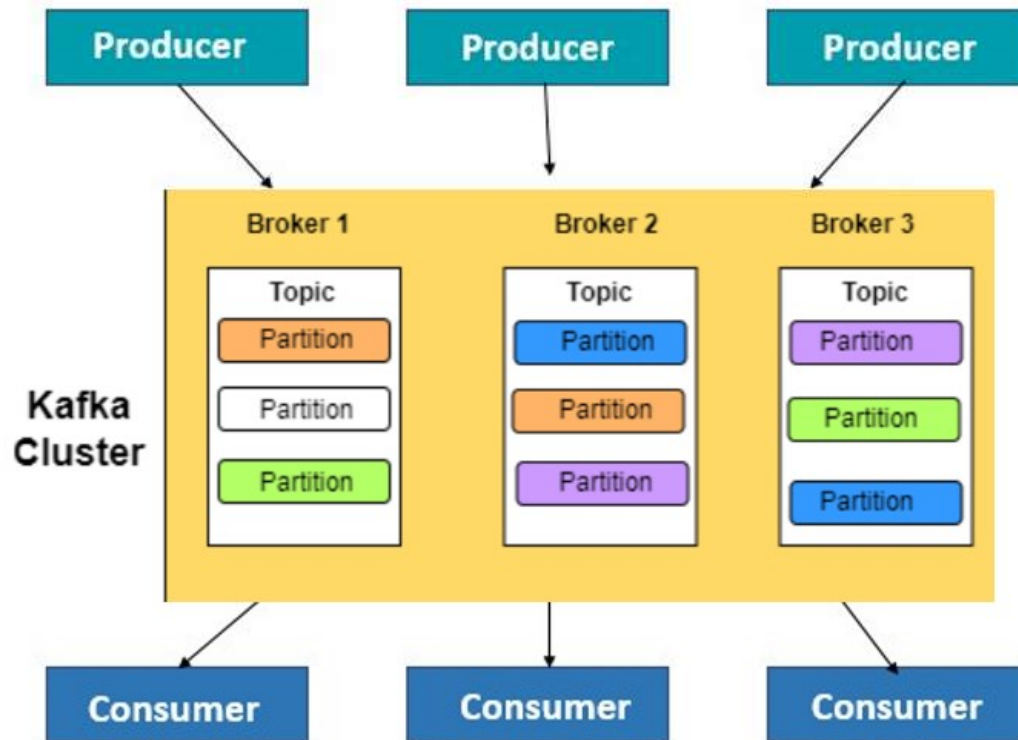


Future Proof NonStop



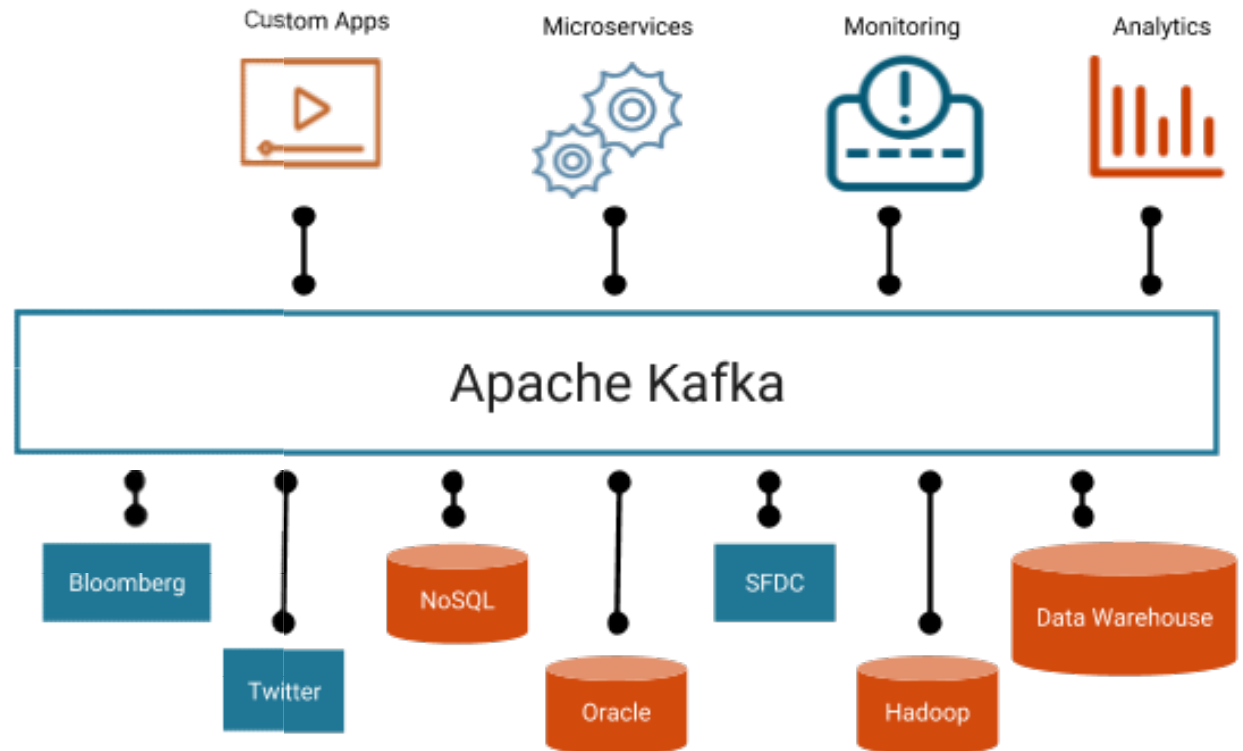
Explore New
Technology

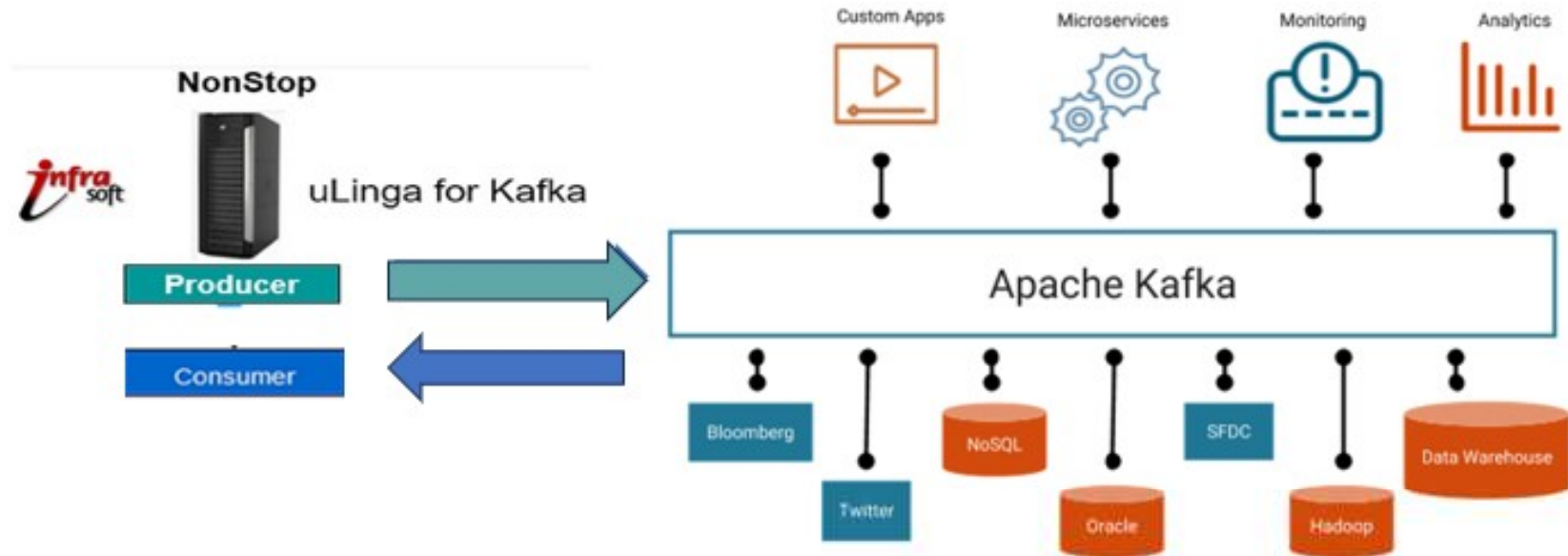
Kafka



Topic	A category of message in Kafka. Analogous to a file or SQL table.
Partition	A Topic can be stored in multiple partition units. Analogous to a file or table partition.
Broker	A server that houses one or more partitions.
Producer	Clients or applications that publish messages to one or more Kafka Topics.
Consumer	Applications that subscribe to one or more Topics and read the messages.
Kafka Cluster	A Kafka cluster consists of one or more servers (Kafka brokers) running Kafka.

Leverage Kafka Connectors



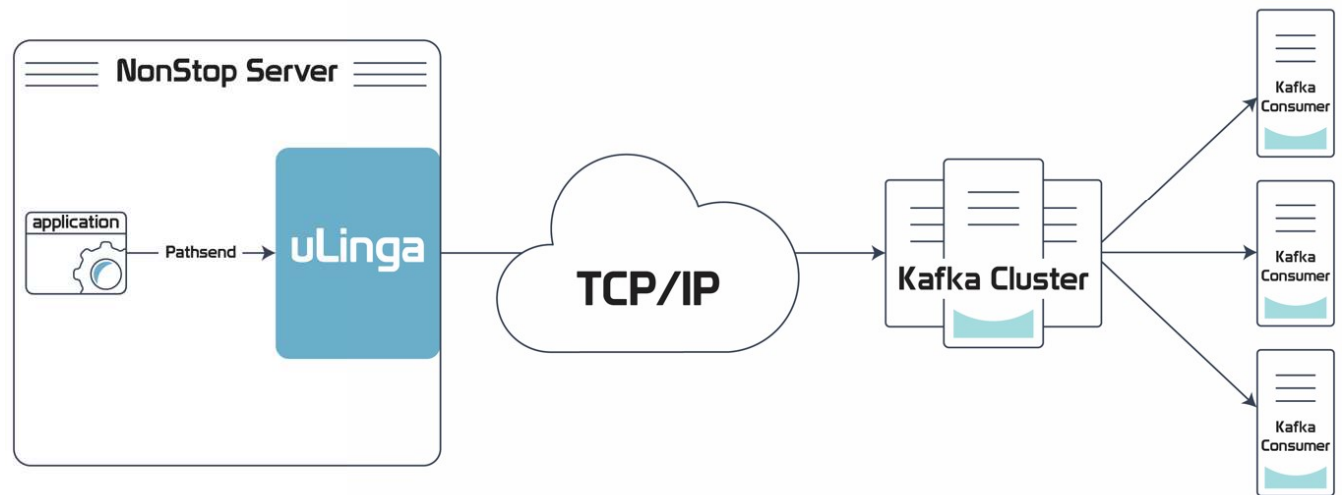


Kafka and NonStop

- Leverage Kafka for NonStop to interoperate with other platforms, software packages and cloud services
- Nonstop as a PRODUCER or CONSUMER using uLinga

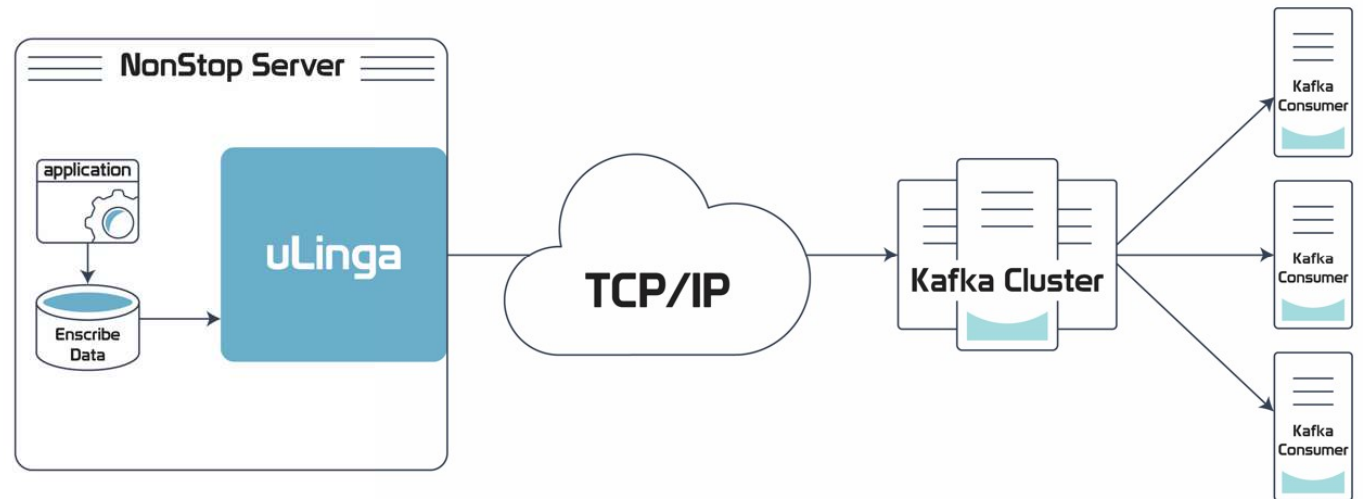
uLinga for Kafka

Pathsend



uLinga for Kafka

File Reader



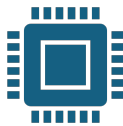
NonStop Use Cases



Publish NonStop application logs or system events to Kafka for real time analysis



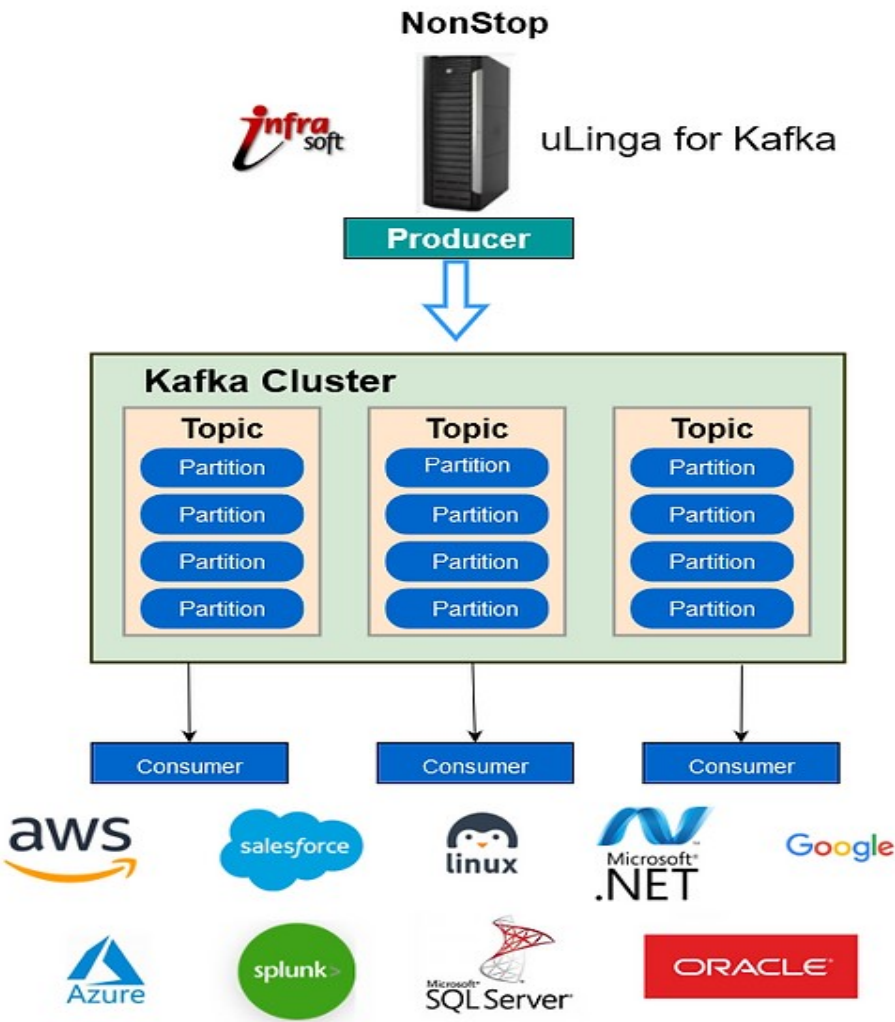
Publish NonStop data to sync with Cloud Services



Push NonStop data changes to other database platforms such as SQL Server or Oracle



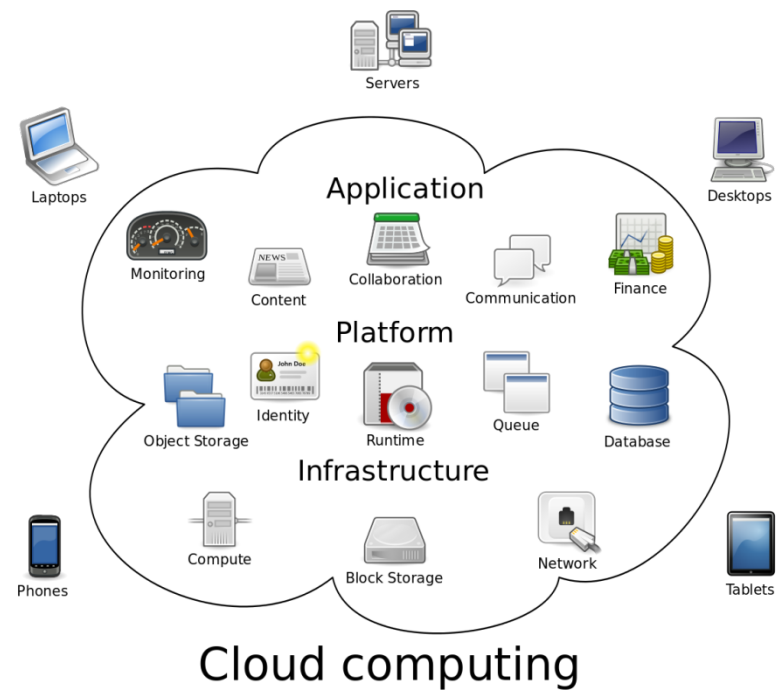
Asynchronous messaging interface to applications on other platforms such as Windows or Linux



Scale out to the Cloud



**HPE
NonStop**



S3utils

Easily move data from your NonStop cluster to Amazon AWS S3 storage.

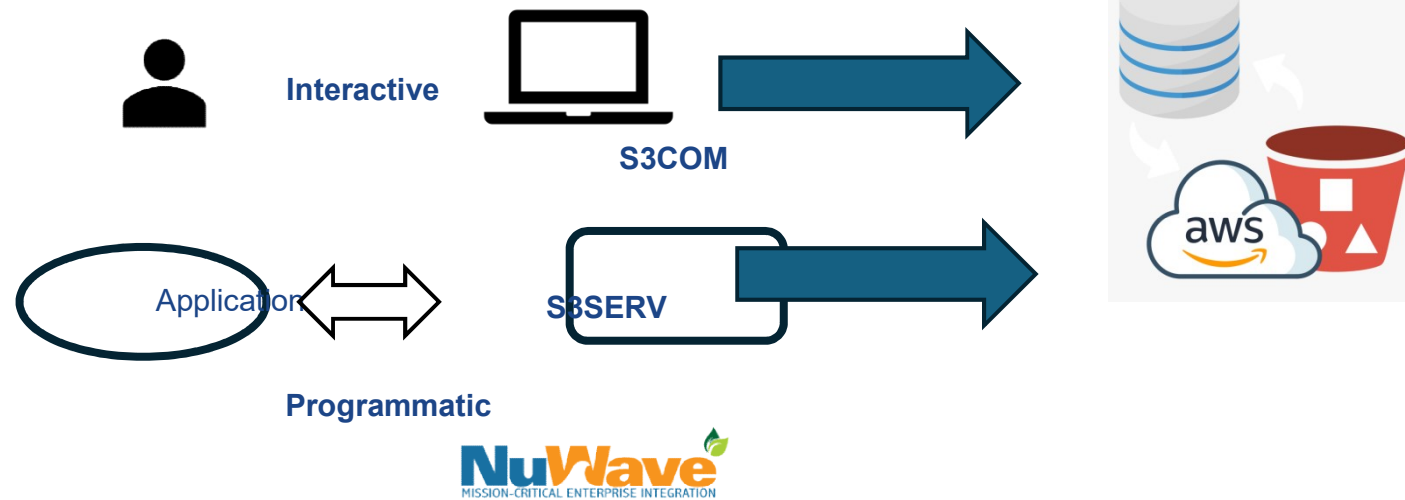
Schedule transfers with NetBatch



Dynamic access to Amazon AWS S3 Buckets

Export near real-time OLTP data easily

S3 Integration Made Easy with S3Utils



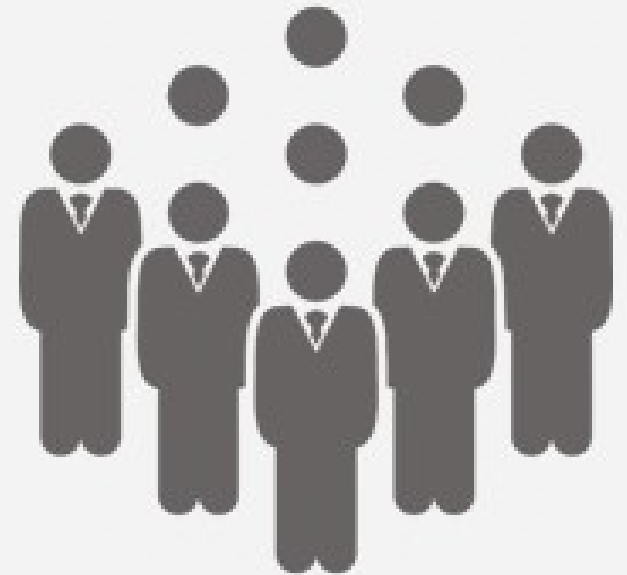
Future Proof NonStop



Invest in people

Challenge

Not enough
trained
resources
on
NonStop



Staff Shortage

Need for Cross Training

Corporate Help Desk

Off shore Support

Task/process oriented

Need to get trained
faster



Training Paradigm Shift

- Just-in-time training
- Training through analogy
- Use GUI tool - TOP



New Approach

- “Just in time” training
 - Content Plan
 - Learn in chunks
 - Apply learning immediately
 - Advance to next level

More productive faster
and better knowledge
retention



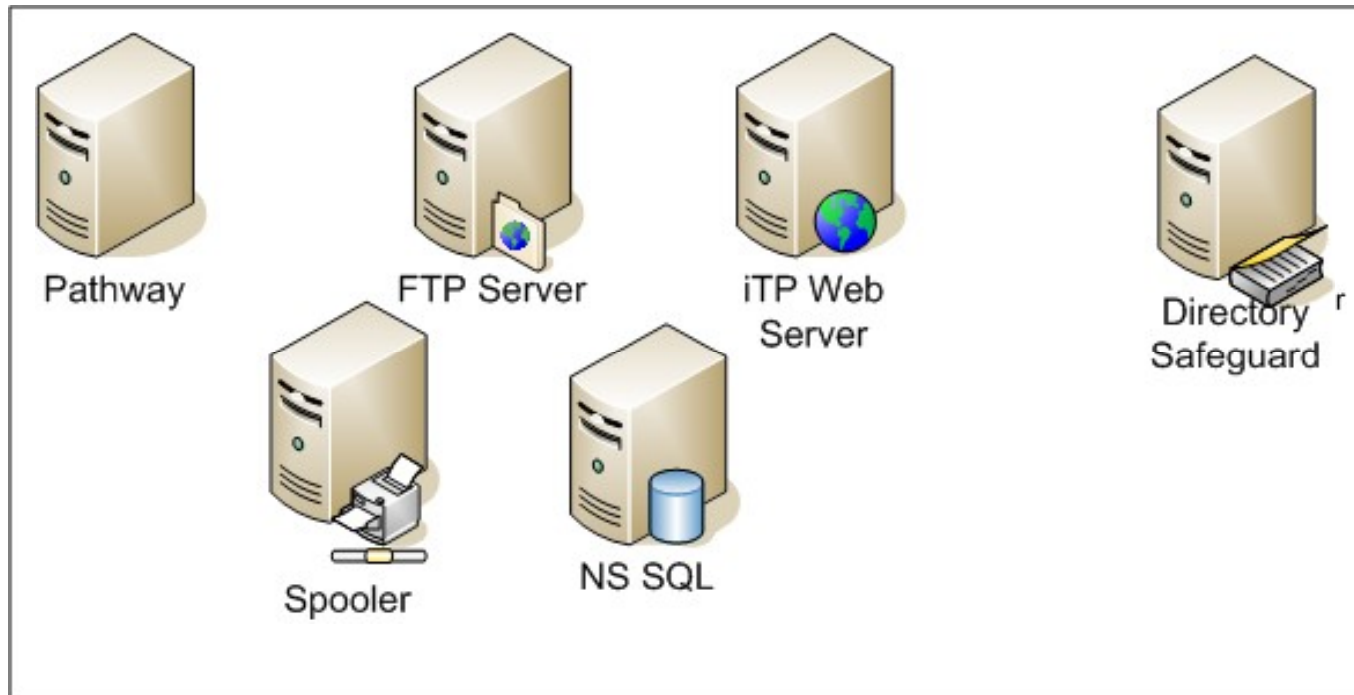
What is NonStop?

- Common answer given:

” NonStop is a scalable, high performance system that has built-in nonstop process pairs and mirrored disks that provide fault-tolerance resulting in 99.9999% uptime.”



What is NonStop?



“ NonStop is a network of servers (like Windows or Linux) that can run different applications written in languages like Java or C, and services (like SQL Database, Web server, Directory) with a built-in load balancing capability like F5.”

Solution

“It is easier for a person to learn NonStop with the right tool and a new training approach”



TIC Fast Path Training



Online Zoom
Training



Lecture



Quiz



Hands-on labs

TIC Fast Path Training Plan

Working with
Files

Working with
Processes

Print
Management

Database

EMS

TACL

Pathway

TMF

TIC can help you



Learn New Technology

NonStop Training



Contact TIC to help you at
e: sales-support@ticsoftware.com
p: 516-466-7990
<http://www.TICSoftware.com>

TIC Software



Right Experience
and Expertise



Right
Solutions



Right Team

Future Proofing NonStop



Don't be complacent



Learn with TIC Webinars



Learn by doing a POC



Collaborate with TIC